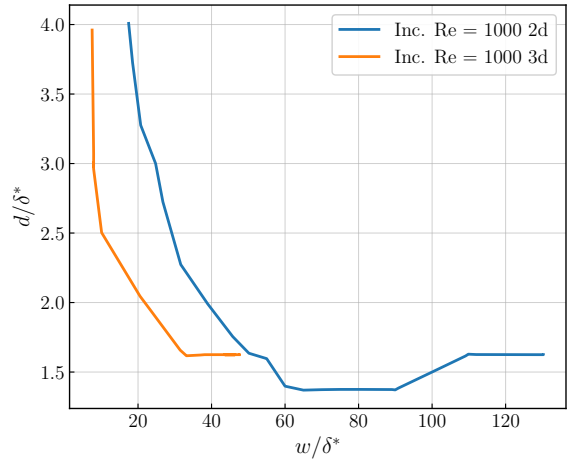


IMPERIAL

incompressible 3d analysis

Víctor Ballester
February 12, 2026

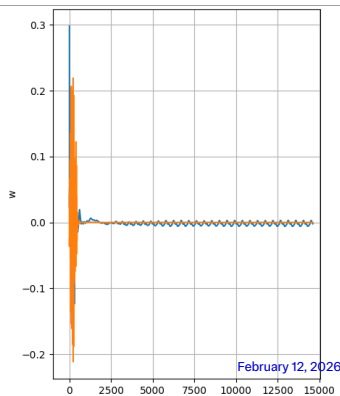
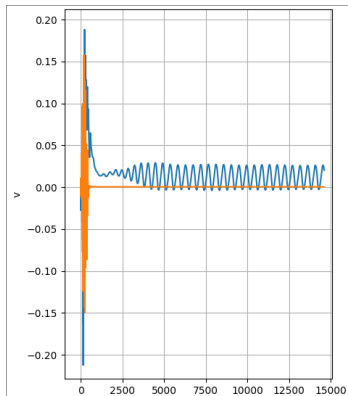
instability curves



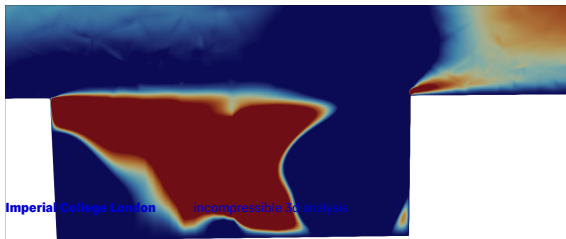
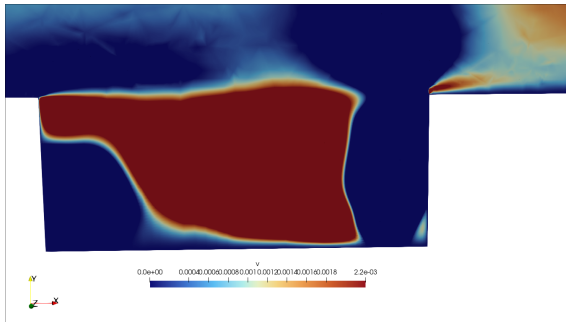
3d instability (in 2d stable regime, $d = 4$ $w = 10$)

- 3d instability is very different from 2d instability.
- For $d = 4$, the unstable mode is fully concentrated in the gap, with very little influence in the boundary layer after the forward facing step.
- Frequency is an order of magnitude lower than the 2d instability:
 $\omega_{w=\text{hopf } 2d}^{d=4} \approx 0.252$ $\omega_{w=\text{hopf } 3d}^{d=4} \approx 0.0136$.

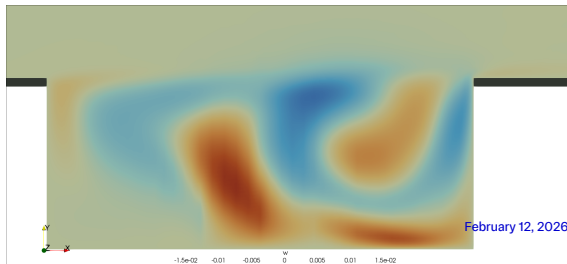
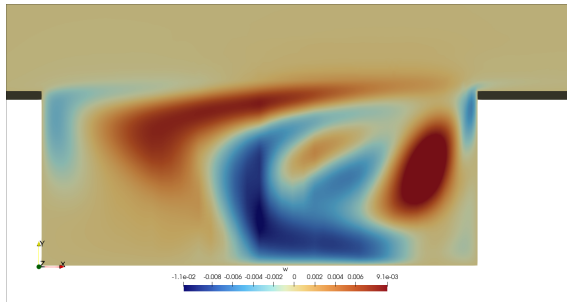
Time series of v and w **inside the gap**
and **in the BL just after the gap**



v at $t = t_1$ (top) and $t = t_2$ (bottom)

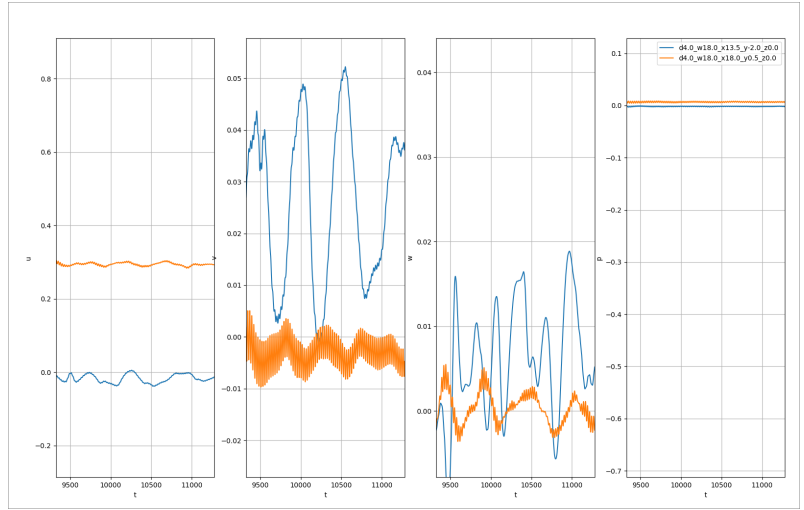


w at $t = t_1$ (top) and $t = t_2$ (bottom)



2d and 3d instabilities coexisting ($d = 4$ $w = 18$)

- Time series of two points **inside the gap** and **in the BL just after the gap** show the coexistence of the 2d and 3d instabilities.
- The 2d instability is only dominant in the BL, not in the cavity.



2d unstable and 3d stable ($d = 1.5$ $w = 50, 80$)

- 3d simulations tend to the 2d unstable mode.
- $\omega_{d=1.5, w=60}^{3d} = 0.142124$
- $\omega_{d=1.5, w=60}^{2d} = 0.1435$

transition at the gap, $d = 1.75$, $w = 80$

